

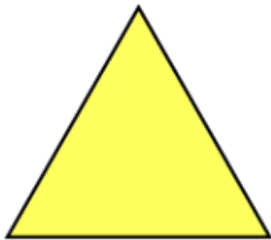
Lesson 2: Perimeter & Area

Friday, May 17, 2024 11:45 AM

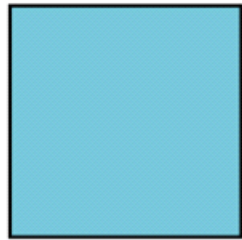
Definition of Perimeter: A perimeter is a closed path that encompasses, surrounds, or outlines either a two dimensional shape or a one-dimensional length. The perimeter of a circle or an ellipse is called its circumference.

Outline the Perimeter of the following Polygons:

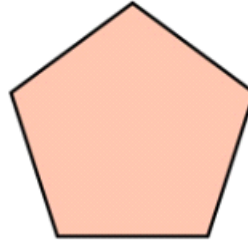
Polygons



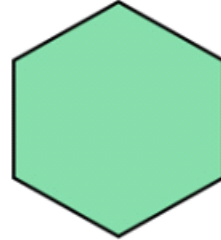
Triangle



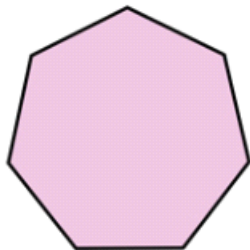
Quadrilateral



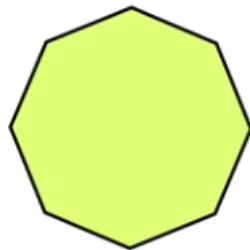
Pentagon



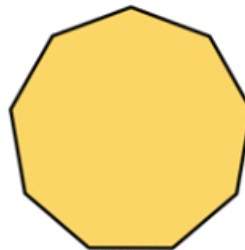
Hexagon



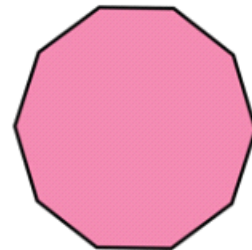
Heptagon



Octagon



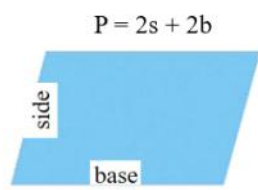
Nonagon



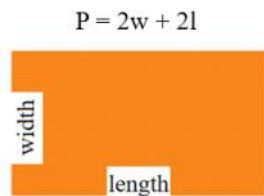
Decagon

Perimeter Formulas & Examples

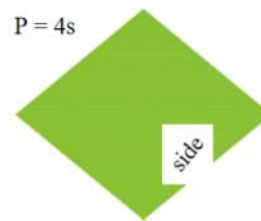
Saturday, June 1, 2024 6:28 PM



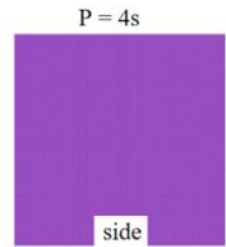
Parallelogram



Rectangle



Rhombus



Square

Many figures can be broken up into rectangles, triangles, and circles.
Therefore, we do not need to memorize a lot of formulas.

And, often, calculating Perimeter is simply a process of adding lengths of sides.

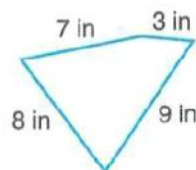
Solve.

1. Find the perimeter of rectangle $EFGH$.

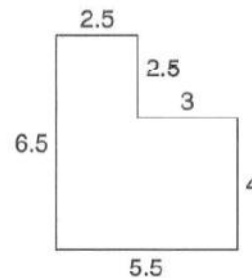


2. One side of a square measures 6.5 centimeters.
What is the perimeter of the square?

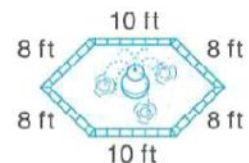
3. What is the distance around the figure?



4. What is the perimeter of the six-sided figure?



5. The fountain in the diagram is enclosed by a low stone wall. What is the length of the wall?



FINDING UNKNOWN SIDES

Example The drawing shows a room from a dollhouse with measurements in inches. What is the perimeter of the room?

Before you can add all the sides, you need to find the missing sides, x and y .

Step 1

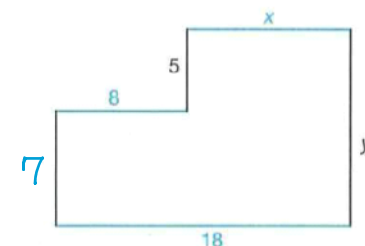
Look at the horizontal sides. The sum of the sides labeled 8 and x is equal to the side labeled 18.

$$8 + x = 18, \text{ so } x = 10$$

Step 2

Look at the vertical sides. The sum of the sides labeled 7 and 5 is equal to the side labeled y .

$$5 + 7 = y, \text{ so } y = 12$$



Step 3.

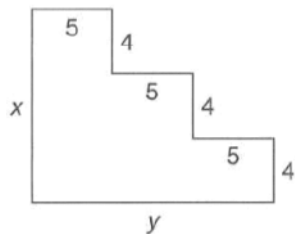
Add the sides.

$$8 + 5 + 10 + 12 + 18 + 7 = 60$$

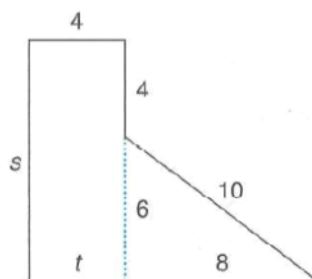
The perimeter of the room is **60 inches**.

Find the perimeter of the following figures.

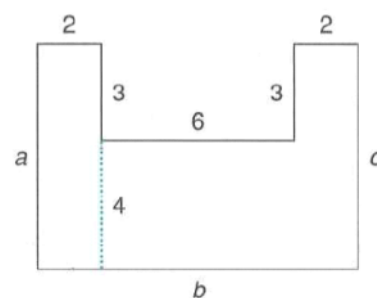
6.



7.

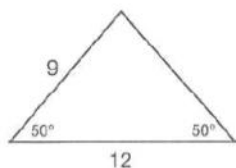


8.

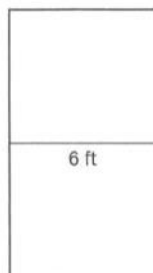


Solve.

9. Find the perimeter of the isosceles triangle.

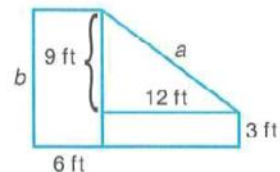


10. The rectangle is two stacked squares. What is the perimeter of the rectangle?



11. **Use Algebra** A rectangular landscape is three times as long as it is wide. If the perimeter of the landscape is 64 inches, what is the length?

12. **Explain** The stage area shown consists of two rectangular platforms and a triangular one. Suppose you need to find the perimeter of the stage. Explain how you could find the measures of sides a and b .



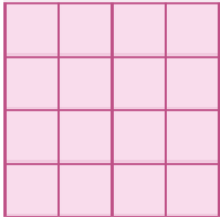
Area

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Definition of Area: an area is the measure of a region's size on a surface.

Finding Area: the surface is divided into squares of equal size. The squares could be, for example, 1 inch by 1 inch, 1 foot by 1 foot, 1 acre by 1 acre, or 1 unit by 1 unit.

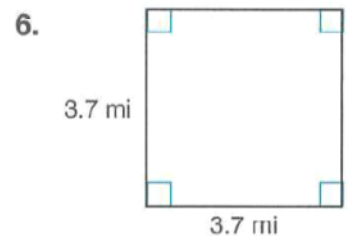
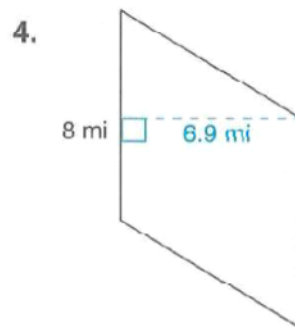
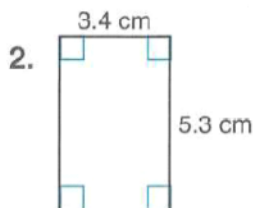
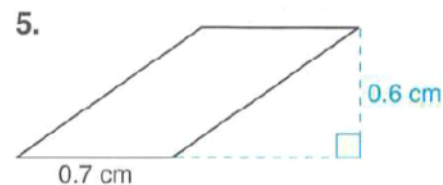
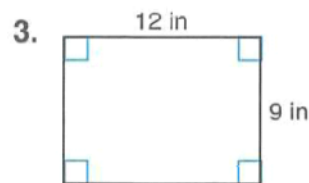
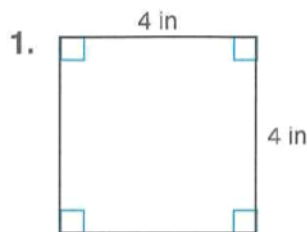
For Example: this surface is $4 \times 4 = 16$ square units.



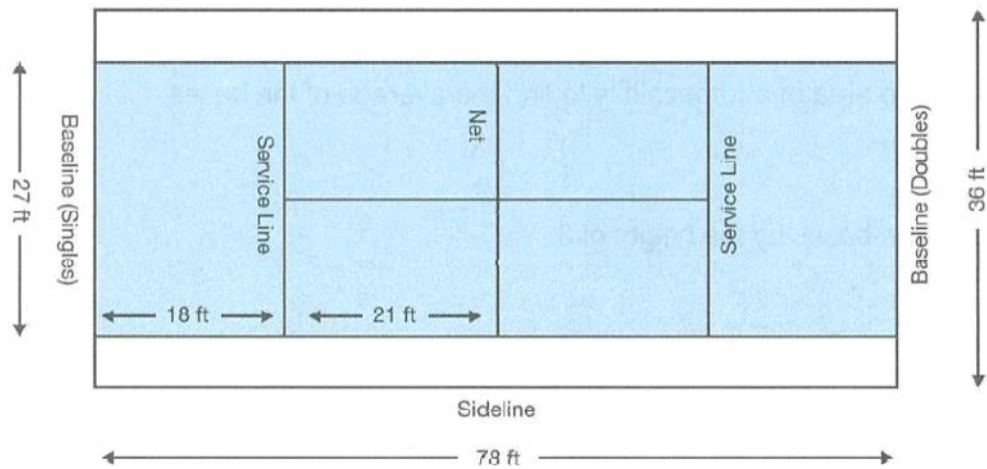
Formula for Area of a Rectangle: $A = l \cdot w = \text{length} \cdot \text{width}$

In the examples below, for #4 & #5, break up the figure.

Find the area of the figures. Remember to label your answer in square units.



Use the diagram of a regulation tennis court for problems 7–10.



7. A singles court will be painted a contrasting color. How many square feet will be painted?
8. On a singles court, what is the area of the rectangle between the service line and the baseline?
9. A ball being served must land in the opposite small rectangle adjacent to the net. What is the area of the space where a serve must land?
10. There are 9 square feet in 1 square yard. What is the area of a regulation doubles court in *square yards*?
11. **Explain** Two rectangles have different dimensions but the same area. Do the rectangles have the same perimeter? Explain your thinking. Draw an example to support your reasoning.
12. **Draw** A figure has an area of 100 square feet. Draw and label a square, a rectangle, and a parallelogram, each with an area of 100 square feet.

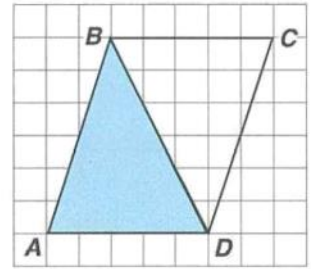
Area of Triangles

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This is a great explanation for the area of a triangle:

Triangles are related to parallelograms. Study the drawing to the right. Compare triangle ABD to parallelogram $ABCD$. What do you notice about the area of the shapes? How are they related?

The area of the triangle is one-half the area of the parallelogram. If you know one, you can solve for the other. Parallelogram $ABCD$ has a base of 5 and a height of 6. Find the areas of both figures.



Area of $ABCD$:	base $\times$ height	$5 \times 6 = 30$ square units
Area of ABD :	base $\times$ height $\div 2$	$5 \times 6 \div 2 = \mathbf{15}$ square units

You can write the following formula based on the example.

Area (A) of a Triangle $A = \frac{1}{2}bh$, where b = base and h = height

Area of Trapezoids

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Look at trapezoid $KLMN$. To find its area, you need to multiply base by height, but a trapezoid has two bases.

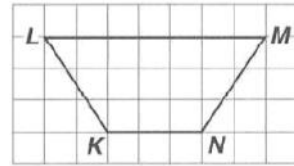
The first step in finding the area of a trapezoid is to find the average of the bases.

$$\frac{LM + KN}{2} = \frac{7 + 3}{2} = \frac{10}{2} = 5$$

Multiply the average of the bases by the height of 3.

$$5 \times 3 = \mathbf{15 \text{ square units}}$$

You are given the formula for finding the area of a trapezoid on the GED math formula sheet.



Area (A) of a Trapezoid

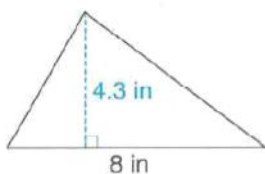
$$A = \frac{1}{2}h(b_1 + b_2), \text{ where } h = \text{height and } b_1 \text{ and } b_2 \text{ are the bases}$$

Triangles & Trapezoids - Example

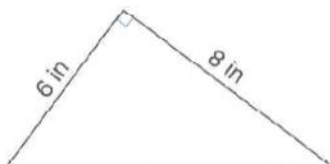
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Find the area of each figure.

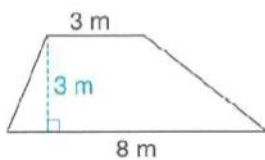
1.



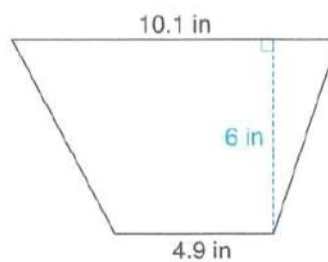
2.



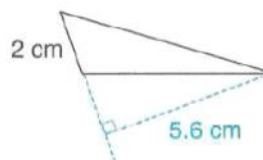
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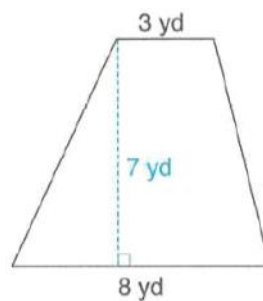
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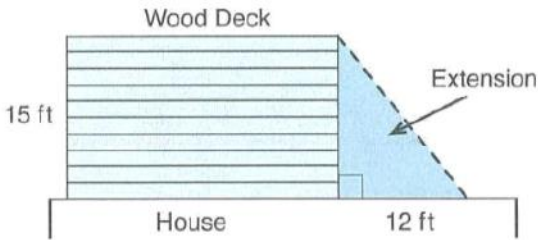


6.



Solve as directed.

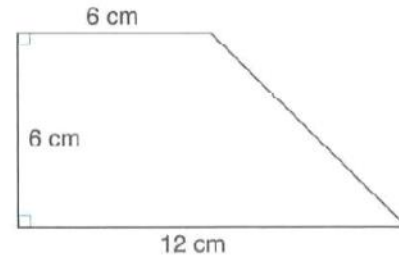
7. Karla wants to add a triangular extension to the deck shown below. How many square feet will be added to the deck?



8. The two legs of a right triangle are 9 inches and 14 inches. What is the area of the triangle in square inches?

9. **Explore** One way to find the area of a figure is to cut it into smaller shapes.

- a. Find the area of the trapezoid using the trapezoid formula.



- b. The trapezoid has been divided into smaller shapes in two different ways. Find the area of the individual pieces and add to find the total area.

